

Exercise- 06

Aim: To write a program to implement error detection and correction using HAMMING Code Concept. Make a test run to input data stream and verify error correction feature.

Code:

```
def hamming_encode (data):
```

```
    d1, d2, d3, d4 = map(int, data)
```

```
    p1 = d1 ^ d2 ^ d4
```

```
    p2 = d1 ^ d3 ^ d4
```

```
    p4 = d2 ^ d3 ^ d4
```

```
    code = f"{p1}{p2}{d1}{p4}{d2}{d3}{d4}"
```

```
    return code
```

```
def hamming_decode (code)
```

```
    bits = list (map (int, code))
```

```
    s1 = bits[0] ^ bits[2] ^ bits[4] ^ bits[6]
```

```
    s2 = bits[1] ^ bits[2] ^ bits[5] ^ bits[6]
```

```
    s4 = bits[3] ^ bits[4] ^ bits[5] ^ bits[6]
```

```
    error_position = s4 * 4 + s2 * 2 + s1
```

```
    if error_position != 0:
```

```
        print ("Error detected at position: ", error_position)
```

```
        bits [error_position - 1] ^ = 1
```

```
        print ("Corrected code: ", "".join (map (str, bits)))
```

```
    else:
```

```
        print ("No error detected.")
```

```
data = f"{bits[2]} {bits[4]} {bits[5]} {bits[6]}"  
return data
```

```
data = input("Enter 4-bit data: ")
```

```
code = hamming_encode(data)
```

```
print("Hamming code: ", code)
```

```
received = input("Enter received 7-bit code: ")
```

```
corrected_data = hamming_decode(received)
```

```
print("Original data: ", corrected_data)
```

input :

1011

Output :

Enter 4-bit data : 1011

Hamming Code : 0110011

Enter received 7-bit code : 0110011

No error detected.

Original data : 1011

Result : Writing and implementing error detection and correction using Hamming code is done successfully. 